



CSE422 Case Study Assignment

**Topic: Discriminatory Use of AI in Law Enforcement
and Predictive Policing**

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Section: 17

1. What does the application in the case study entail? Provide a detailed summary of the application, and provide references to papers with similar applications if you find any

Answer: The case study discusses the practical use of Artificial Intelligence (AI) in law enforcement, including predictive policing and risk assessment systems. Predictive policing uses algorithms to examine previous crime data and anticipate where future crimes are likely to occur or who may be involved in criminal activity. **According to Lum & Isaac (2016)**, this is accomplished by processing enormous volumes of data, such as geographic information, past crime reports, and social aspects such as demographics or socioeconomic status. Predictive tools aim to optimize the deployment of law enforcement resources by predicting crime patterns, reducing response times, and preventing crime in high-risk areas. One prominent application of predictive policing is the COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) tool. **ProPublica (2016)** claims that COMPAS generates a score based on variables like age, family history, and criminal history in order to evaluate the probability of recidivism. It has been frequently utilized in the criminal justice system of the United States to determine sentencing and parole. However, COMPAS and other predictive tools have been criticized for racial bias. **According to ProPublica (2016)**, white defendants with comparable criminal histories were frequently categorized as lower risk, while Black defendants were disproportionately identified as high-risk for reoffending by COMPAS, even when they did not commit further offenses. This shows how algorithmic tools can unintentionally create systemic inequality in the justice system. Similar predictive policing applications have been deployed in multiple cities worldwide. According to Shroff (2019), programs like PredPol, which predict crime hotspots, make claims about lowering crime and increasing the effectiveness of law enforcement. But these systems can reinforce biases present in historical arrest data, giving minority or low-income communities. In the end, these AI technologies show both the potential to improve police effectiveness and the possibility of unintentionally reinforcing social bias, underscoring the necessity of strict supervision and ethical thought in their use.

2. What are the ethical implications of the application mentioned in the case study? Is there any other paper that talks about problems related to a similar domain?

Answer: The use of AI in law enforcement has significant ethical implications, especially when it comes to risk assessment and predictive policing. Despite being intended to increase productivity and lower crime, these instruments frequently raise serious concerns about accountability, transparency, and fairness.

- ☐ **Bias and Discrimination:** Predictive policing algorithms are typically trained on historical data, which can contain embedded biases. **O'Neil (2016)** says that high arrest rates and a history of overpolicing, especially in minority neighborhoods, increase the likelihood that an area will be designated as high-risk. This strengthens racial disparities in the criminal justice system and encourages the cycle of excessive policing. In particular, **Angwin et al. (2016)** revealed that COMPAS consistently overestimates Black

defendants' likelihood of reoffending while underestimating that of white defendants, illustrating how AI can exacerbate institutional racism.

- ☐ **Lack of Transparency:** A large number of AI-powered policing technologies are closed-source and private. Neither the general public nor law enforcement officials fully understand how these systems operate, which makes it challenging for people to contest possibly unfair or discriminatory results (**Angwin et al., 2016**). Ethical questions about responsibility in the legal system are brought up by this lack of transparency.
- ☐ **Privacy and Human Rights:** Predictive policing requires collecting and analyzing large volumes of data, including sensitive personal information. **Eubanks (2018)** explains that the widespread surveillance enabled by AI tools can disproportionately impact marginalized communities, raising concerns about privacy and civil liberties.

In conclusion, although AI technologies in law enforcement are effective and predictive, they also pose significant ethical hazards, including bias, a lack of transparency, and privacy dangers, which need to be properly addressed in order to prevent systemic injustices.

3. If such an application is deployed on a global scale, what could be the long-term future impact on society? Provide your opinion on this, and if possible, provide suitable references to support your argument

Answer: Global deployment of predictive policing tools like COMPAS and PredPol could have profound consequences

- ☐ **Increase of Social Inequalities:** According to **Brantingham & Brantingham (2008)**, artificial intelligence (AI) technologies that are based on past crime data have the potential to perpetuate current disparities. So, worldwide deployment may systemically harm underprivileged areas, resulting in protracted cycles of overpolicing and social disadvantage.
- ☐ **Erosion of Public Trust:** Biased predictive policing may cause social unrest and resistance by eroding public trust in law enforcement, particularly in areas that feel unfairly singled out. (**Edwards & Veale, 2017**). As a result, communities unfairly targeted may lose confidence in law enforcement, reducing cooperation and potentially triggering social unrest
- ☐ **Privacy and Surveillance Concerns:** **Eubanks (2018)** says that the widespread use of predictive policing may result in universal monitoring systems where people's whereabouts and actions are continuously tracked, which would raise grave issues about civil liberties. So basically, it creates a "Big Brother" scenario, constantly monitoring personal movements and associations.
- ☐ **Legal and Policy Challenges:** **Edwards & Veale (2017)** note that global deployment would create complex debates over regulation, oversight, and fairness, as countries with weak oversight may misuse these systems while others may enforce strict controls. This creates disparities in justice outcomes.

In conclusion, while AI can improve law enforcement, if not carefully regulated, it might reinforce inequality, jeopardize human rights, and erode public confidence in the rule of law.

Reference

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